

Principles of Programming Languages

Lecture 12: Paradigms: Logic Programming.

Andrei Arusoaie¹

¹Department of Computer Science

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Outline

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- ▶ Yes: Logic programming

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- ▶ Proof strategies: *goal-directed* vs. *forward-reasoning*
- ▶ Logic programming uses *goal-directed*: for the current goal, search the rule that might have been applied to arrive at this conclusion

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- ▶ Logic programming can be seen as controlled deduction
- ▶ Logic programming languages (or families of PL): Prolog, ASP (Answer set programming), Datalog

Logic programming: how it works

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- ▶ Backtracking: more than one possible rule applicable
- ▶ Subgoal order: which of the premises to consider?

Demo

Prolog demo: light meal problem

Resources

Reading material:

- ▶ **Chapter/Lecture 1:** <https://people.mpi-sws.org/~dg/teaching/lis2014/modules/lp-fp-07.pdf>